

PRATIMAI · AI HANDOUTS

AI Around Me

My detective notebook for spotting smart machines

CLASS 6 · LEVEL 1 — NOTICE

20 PAPER MISSIONS · 5 CHAPTERS · NO SCREEN NEEDED

PRATIMA AI Curriculum Team

How do I know? · What is missing? · Who made this?

THIS CASE FILE BELONGS TO

_____

name · class · roll number

CASE OPENED ON ★ CASE CLOSED ON

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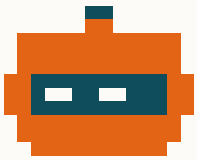
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Set in **Nunito** and **Baloo 2** on warm paper stock. Both typefaces are used under the SIL Open Font Licence. Every chart, grid, diagram and pixel drawing in this book is drawn as crisp vector art, so each page prints cleanly at any size.



PRATIMAI · AI HANDOUTS

Acknowledgments

To the teachers and their Classes 6 and 7 who piloted every mission with pencils, paper and great patience — your classroom notes shaped this book. You told us which tasks made the room go quiet with thinking, which ones made it erupt in argument, and where the instructions needed one more example. Every one of those notes made a page here better.

This series was built brick by brick by three teams working as one. The **AI research team** checked every explanation against how machines really learn, so nothing in this book is a myth wearing a friendly font. The **academic team** walked every chapter beside the school syllabus, so each game quietly trains a skill that examinations will ask for later. And the **child-psychology team** shaped the pace of every page: short missions for young attention, quick wins that open every chapter, confidence circles that make honesty feel safe, and space to write on every task — because an idea worked by the hand is remembered by the mind. Alongside them, curriculum reviewers guarded the "unplugged" promise: not one mission in this series asks for a phone, an account or a password.

Thanks go as well to the designers of Nunito and Baloo 2, whose friendly letterforms keep pages of hard thinking feeling light. And finally, to the detective holding this book: every confidence circle, empty line and doodle box in here was left blank on purpose — because the last author of this case file is you.

20 PAPER MISSIONS · 14 WORDS TO KEEP · 5 CHAPTERS · 1 BADGE EARNED

How to Use This Handbook

Welcome, Detective! This book is your **case file** for the biggest mystery of our time: machines that seem smart. You will not need a computer, a phone or an app — your tools are paper, pencils, cards and your own sharp eyes. Every chapter hands you missions (we call them **tasks**), strange clues to examine, and empty spaces to fill with what you discover. Real detectives write everything down, so this book belongs to you: write, draw, tick, cross out and try again.

WHY THIS BOOK WORKS THE WAY IT DOES

Your brain learns by **doing**, not by reading about doing. That is why every chapter opens with a quick win, explains only after you have tried, and asks you to **write** — writing pulls an idea from half-known to known. The confidence circles are not decoration: asking **how sure am I?** is a real thinking skill (psychologists call it metacognition), and it is the same habit scientists use. Missions are short because attention works in sprints. Ideas return in later chapters on purpose — like meeting a stranger twice, until they feel like a friend. And nothing here can be failed: a wrong answer you go on to check is just a clue you found early.

1 READ

Read the mission twice. Underline anything that puzzles you — puzzles are clues.

2 DO

Do the task with your partner, group, class or family, exactly as the labels say.

3 WRITE

Write what you saw in the spaces, then shade your confidence circles honestly.

The work-mode labels

ALONE	your own case — you investigate by yourself
PAIR	two detectives — share one book, take turns reading codes aloud
GROUP	three to four detectives — decide roles before you start
CLASS	the whole class plays together, led by your teacher
HOME	a mission for home — show a grown-up what you discovered

HOW TO READ A MISSION

T6-01 Mission title — **PAIR · 10 MIN**

★ **QUICK WIN** sits at the right edge of the easier missions.

Every mission has a code like **T6-01** — **T6** means this Class 6 handout, and **01** is the mission number, so your teacher can say “open Task T6-08”. The **PAIR** label tells you who to work with, **10 MIN** is your time limit, and **QUICK WIN** marks an easy first success. Chapters 1 and 2 both start with a Quick Win to warm up your detective muscles — nobody fails in this book, because every attempt teaches a clue.

THE QUESTION HABIT — YOUR SUPERPOWER

Three questions make you a sharper detective than any machine: **"How do I know?"** · **"What is missing?"** · **"Who made this?"** Use them in this book, on the internet, in the news, and when a machine gives you an answer that looks too perfect. Detectives are not people who know everything — they are people who **keep asking**.

PRACTISE THE THREE QUESTIONS — RIGHT NOW

Your friend shows you a video: "A robot in Bengaluru wrote a poem all by itself!" Hold each question up to the claim, one at a time:

1. **How do I know?** — What would prove the robot wrote it alone?

2. **What is missing?** — What does the video NOT show?

3. **Who made this?** — Who built the robot, and who posted the video?

A Note for Grown-Ups

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 20 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. The best help you can give is to ask the three Question-Habit questions, praise careful **reasoning** more than quick right answers, and treat the confidence circles as honest thinking, not scores. A separate teacher pack carries answer notes and rubrics; this book deliberately never prints solutions beside a task.

Four ways to help — without giving answers

1 Ask, don't tell

When your detective is stuck, ask: **"What have you tried? What does the machine know — and how would it know that?"** A good question unblocks; a quick answer un-trains.

2 Praise the reasoning

Swap "correct!" for **"I like how you checked that"** or **"you changed your mind when the evidence changed — that is real detective work."**

3 Let it be wrong

A confident wrong answer is useful material. Ask **"how sure are you — and what would change your mind?"** Changing a belief is the skill this book teaches.

4 Connect to real life

Point at the machines around you — the washing machine, the map app, the shop's barcode — and ask which ones **follow steps** and which ones **learned from examples**.

IF YOUR CHILD ASKS…

"Is the machine actually thinking?" — Say: it is finding patterns in examples, the way you learned what a mango is. It does the pattern part fast; **you** do the judging part better. **"Can it be wrong?"** — Say: always possibly. That is why every mission here ends with **how sure am I?** — the same question scientists, judges and pilots use. **"Is AI dangerous?"** — Say: tools take after their makers. That is why this book keeps asking **who made this, and who is responsible?** — the question that keeps people in charge.

SIGNS OF DETECTIVE GROWTH WORTH PRAISING

Watch for these quiet changes — they matter more than any correct answer: your child **asks how sure** they are before answering · they **check a second source** without being told · they **change their mind** when the evidence changes — and say why · they explain a machine's **limit**, not just its magic · they teach the three questions to a younger sibling. When you spot one, name it: "That is exactly what detectives do."

My AI Journey Map

THE FIVE STRANDS YOU WILL MEET AT EVERY STOP

U	D	L	W	R
Understand AI	**Data**	**Learn**	**World**	**Responsibility**
What is AI — and what is it NOT?	What do machines learn from?	How do machines get better?	Where is AI hiding near me?	How do I stay safe and fair?

THINKING-TOOLS THREAD

runs through every stop: **exact steps** · **patterns** · **if-then rules** · **how sure am I?**

THE FIVE STOPS ON YOUR CASE

1. **Machines That Seem Smart** — *rule-followers vs learners*
2. **Data: The Food of AI** — *numbers, words, pictures, sounds*
3. **Patterns & Decisions** — *if-then trees · how sure am I?*
4. **Be a Smart Digital Citizen** — *privacy · passwords · fakes*
5. **My Neighbourhood AI Investigation** — *your own investigation*

BY THE END OF THIS BOOK, I CAN…

- [ ] say what AI is — and what it is not — using examples from my own day
- [ ] name two things people do better than machines and two things machines do better
- [ ] name the four kinds of data and sort them into a tidy table
- [ ] turn a simple picture into numbers on a grid, like a real camera does

- [ ] find a pattern, predict what comes next, and say **how sure** I am
- [ ] describe three ways machines learn: labelled examples, grouping, trial and reward
- [ ] point to AI at home, at school and in my neighbourhood, and name the data it uses
- [ ] use my five safety habits: keep private things private, strong passphrases, think before posting, be kind, check before believing
- [ ] explain why an AI answer can be wrong — and how I would check it
- [ ] break a task into exact steps that another person can follow

MY STARTING LINE — a pledge to myself

2 MIN

Fill this in **before** Mission 1 — you will read it again on the last page, and see how far the evidence took you.

One thing I already know about smart machines:

One question I want this book to answer for me:

One skill I promise to practise honestly — no copying, no skipping the write-ins:

Signed: _____ Date: _____

MY QUESTION CARD — cut it out, keep it in your pencil box

1. How do I know? **2. What is missing?** **3. Who made this?** **4. What is the trick here?**

Any time a machine — or an advertisement, or a forward — sounds too smart, the card comes out. Detectives are not people who know everything; they are people who keep asking.

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The boxes in this book

You will meet these boxes on every case. Each one wants something different from you — collect them all.

MISSION

A paper mission with a code like **T6-01**. The label tells you who you work with and how many minutes you get. Do it — then write. Writing **is** the mission.

WORD POWER

A word worth keeping in your detective kit. Read it three times today — words are the tools of thinking.

SELF-CHECK

Skills to grade yourself on. Shade a confidence circle honestly — honest circles make you learn faster.

THINK & INK

A question with lines. There is no single right answer here — your **reasons** are the answer.

Also on patrol: **MYTH BUSTER** busts a wrong idea many people believe · **DETECTIVE'S NOTE** hands you a professional's secret · **HOME LINK** (dashed) is a mission to take home and teach a grown-up.

1. Machines That Seem Smart

GUIDING QUESTION

When is a machine actually learning — and when is it only following steps?

Every detective starts by learning to tell two suspects apart. In this chapter you will meet **rule-followers** — machines that run exact steps — and **learners** — machines that improve from examples. By the last mission you will have played a human robot, taught an alien using only flashcards, and judged a fair humans-vs-machines contest.

MISSIONS T6-01 – T6-05
OUTCOMES 6.U1 · 6.U2 · 6.L2 · 6.W1 · 6.T1 **STRANDS** U W

Links: Maths — patterns · English — exact instructions

In this chapter

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WORDS YOU'LL MEET HERE

algorithm **automation** **artificial intelligence** **training examples** **predict**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Write the name of one machine you believe is smart — and the one thing it did that impressed you. You will re-read this line after Chapter 5, at the end of the case, and see if your idea changed.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · eraser · 10 blank cards · a partner's sharp eyes

1.1 Smart machines are everywhere

Every day, machines seem to do clever things. A video app picks the next clip you actually want to watch. A voice assistant answers when you ask about the weather. A map app finds a faster road when traffic jams up. It is easy to think: **the machine is smart, like a person.** But a good detective looks closer before deciding. Under every clever trick there are only two possible explanations: either somebody wrote **exact steps** for the machine to follow, or the machine **learned a pattern from examples**. On this case, you will learn to tell the two apart — and that one skill will change how you see every screen, speaker and gadget around you.

T6-01 Human Robot

QUICK WIN · PAIR · 10 MIN

One of you is the **Robot**, the other is the **Engineer**. The Engineer writes exact steps for the Robot to make a paper plane, or to walk from the door to the chalkboard. Then swap. The Robot must follow the steps **literally** — if the step says “fold the paper”, fold it any way you like! When both of you have played both roles, write here:

One step my Robot followed too literally:

What this taught me about giving instructions to machines:

Our funniest too-literal moment:

WORD POWER · 1

Definition: **algorithm**

An exact list of steps that tells a machine how to do a task, one step at a time.

Read it three times today — it sticks!

DETECTIVE'S NOTE

The Human Robot game is why programmers and AI designers must be **exact**. A machine never guesses what you **meant** — it only knows what you **said**. When a machine does something surprising, ask: was the algorithm wrong, or was it the data?

1.2 Rule-followers: same steps, every time

A washing machine washes exactly the same way today as it did last month. A lift goes up and down by fixed rules. A calculator will never get better at adding — it already follows its algorithm perfectly. Machines like these do **automation**: work done by fixed steps, repeated the same way every time. They are useful and fast, but notice something important: they never **improve**. Nobody has to teach them; nobody can. They are rule-followers, and rule-followers are not the same as learners.

T6-02 Rule or Learner?

PAIR · 10 MIN

Sort these eight machines. For each one, write **R** if you think it mostly follows fixed rules, **L** if you think it learns from examples, or **?** if you are honestly not sure. Then write one clue that helped you decide.

Machine	R / L / ?	One clue that helped you
"Washing machine"		
"Lift"		
"Video app that suggests clips"		
"Spam filter in email"		
"Calculator"		
"Traffic signal"		
"Voice assistant"		
"Map app that reroutes you"		

The machine I argued about most with my partner:

WORD POWER · 2

Definition: automation

Work done by a machine that repeats fixed steps in the same way every time.

Read it three times today — it sticks!

AT A GLANCE · RULE-FOLLOWER VS LEARNER

Ask yourself…	Rule-follower	Learner
Where do its steps come from?	A person wrote the algorithm.	It found the pattern in training examples.
What happens with a brand-new case?	It does the same steps anyway.	It makes a prediction from what it learned.
Can it get better at its job?	Never — until a person changes the steps.	Yes — more examples can improve it.
One example from my day		

MYTH BUSTER

MYTH *"AI is a robot."*

TRUTH AI is a **program** — a learner made of data and patterns. It can live inside a phone, a website, a car or a power plant, and it has no body at all. A robot is one place a program can be kept; plenty of robots run with no AI, and plenty of AI runs with no robot.

60-SECOND HUNT — DO IT BEFORE YOU READ ON

Look up from this page. Point at one **rule-follower** and one **learner** within three metres of you — then defend your choices to your partner in one sentence each. Disagreeing is allowed: arguing from evidence is exactly the skill this chapter teaches. Write your two choices:

Rule-follower:

Learner:

1.3 Learners: machines that improve from examples

So how does a video app get good at suggesting clips you like? Nobody wrote a rule for "show Aarav cricket clips". Instead, the app **learned from examples** — millions of them. It noticed which clips people similar to you watched, and it found **patterns** in that. This is the big secret of this whole handout: a learning machine is not given the answer; it is given **examples**, and it works the pattern out by itself.

Here is the strange part: the people who build AI do not write the final rules either. **They choose the examples and the data**, and the machine does the pattern-finding. That is why the examples matter so much — feed a learner one-sided examples, and it learns a one-sided view of the world, exactly like the alien in your next mission.

T6-03 Teach the Alien

GROUP · 15 MIN

In your group of four, one person is the **Alien** — smart, but knowing nothing about Earth. The other three must teach the Alien the idea of **mango** using ONLY examples: point at real objects, drawings or the cards your teacher gives you, and say "mango" or "not mango". No describing words allowed — aliens don't speak Human! When the Alien starts guessing correctly, play the twist round: show **only green mangoes**. Then answer:

What wrong idea did our examples put in the Alien's head?

How could we choose better examples next time?

WORD POWER · 3

Definition: **artificial intelligence (AI)**

Technology that lets machines do tasks that seem to need human intelligence.

Read it three times today — it sticks!

WORD POWER · 4

Definition: **training examples**

The examples we show a machine so that it can find patterns by itself.

Read it three times today — it sticks!

DETECTIVE'S NOTE

Machines **predict, match and estimate** — those are the verbs of AI. People **design** the system, **choose the data** and are **responsible** for what it does. Keep those verbs straight and you will never be fooled by a "smart machine" story.

1.4 Humans vs machines — the fair contest

So is a machine ever really smart? Try a fair contest. Machines are dazzling at some jobs: counting a million numbers without a mistake, noticing a pattern in a million examples, working all night without getting bored. People are dazzling at others: understanding a joke, knowing that your friend is sad from one look, deciding what is **fair**, asking a question nobody has asked before. The contest has no overall winner — and that is exactly why people and machines make a good team, as long as the **people** stay in charge of the judging.

T6-04 Better At / Worse At

ALONE · 10 MIN

Fill the table, then your class votes for the most surprising idea.

Two things PEOPLE do better — and why	Two things MACHINES do better — and why

The class's most surprising idea was...

HOME LINK *take this page home — no screen needed*

Ask a grown-up: "Which one job would you NEVER hand to a machine — and why?" Write their reason, then your smartest counter-question:

HOME LINK *take this page home — no screen needed*

Ask a grown-up: “Which one job would you NEVER hand to a machine — and why?” Write their reason, then your smartest counter-question:

HUMAN VS MACHINE **One example or a thousand?** HUMAN AI

Here is the fair contest’s second round — how your brain and a machine learn the very same thing differently.

THE HUMAN WAY

You can learn “mango” from ONE mango — and you can still spot a blue-painted mango as a mango. Your brain mixes sense, story and experience together.

THE MACHINE WAY

The machine needs MANY labelled examples. Show it only green mangoes and a blue one confuses it — it holds the pattern, not the fruit.

One thing I could learn from a single example:

One thing a machine can do with thousands of examples that I could never do:

Ask a grown-up: which one is smarter? Write why the honest answer is “both — in different ways”:

1.5 Three ways machines learn — three little stories

Machines learn in more than one way, and each way has a human story you already know. Watch for all three in this book.

STORY 1
Learning with labels
Show it, tell it. Like learning fruit names with flashcards: your teacher shows a card and says “mango”. The machine gets examples **with** answers attached — “this photo is a cat”, “this one is not”. It learns to match new photos to the labels.

STORY 2
Grouping without labels
Sort it your way. Like tidying a mixed toy box with no instructions: you make your own piles — colours here, shapes there. The machine gets examples with **no** answers and must find its own groups, like sorting customers or animals it has never been told about.

STORY 3
Trial and reward
Try it, score it. Like learning to ride a bicycle: wobble, fall, adjust, try again — and a happy feeling keeps you going. The machine tries actions, gets points for good ones, and slowly learns the moves that earn the most reward.

Real AI systems often mix all three. Class 7 will show you what each way is called — for now, the stories are enough.

DETECTIVE DRILL · WHICH STORY IS IT?

The learner…	Story 1, 2 or 3?
Leena learns dance by copying her teacher’s moves, with a clap for each correct step.	
Irfan tidies a mixed pencil box by making his own piles — sharp ones here, colours there.	
A baby learns “dog” because everyone points and says “dog!” at every dog.	
A cricketer improves by trying shots, missing, adjusting — and loving the boundary!	

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I can point to a **rule-follower** and a **learner** in my own home, and say which is which ○○○
- ☐ I can explain “learning from examples” in my own words, without the book’s help ○○○
- ☐ I can name one thing people do better than machines — and say **why** ○○○
- ☐ I can retell the three learning stories and match a new example to the right story ○○○

THINK & INK

One machine I use seemed really smart. Now I think it is a … (rule-follower / learner), because …

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I thought "smart machine" meant one thing. Now I know there are **two kinds**:

MYTH BUSTER

MYTH *"AI is alive — it has feelings."*

TRUTH A learner finds patterns in numbers; it does not **feel** anything. When a chatbot writes "I am happy to help!", that sentence was predicted from patterns in text people wrote. The **people** who design, choose data and take responsibility have the feelings — and the responsibility.

TWO MACHINES, TWO VERDICTS — THE 30-SECOND SORT

A fridge that beeps when the door stays open

rule-follower or learner — and how do you know?

A keyboard that predicts your next word

rule-follower or learner — and how do you know?

HOME LINK *take this page home — no screen needed*

T6-05 AI Detective at Home

HOME · 15 MIN

Walk through your **morning routine** like a detective. At each moment below, tick whether a machine with possible AI **may** hide there, and tick whether you think it **learns** from examples or just follows rules. Ask a grown-up if you are stuck — explaining your guess is part of the mission.

Moment in the morning	Possible AI? (✓)	Learns or follows rules?	What data might it use?
"Waking me up (alarm / phone)"			
"Brushing (electric brush / tap)"			
"Breakfast (microwave / kettle)"			
"Checking the day (weather app)"			
"Journey to school (bus / car / map)"			
"Buying something (shop / UPI / ATM)"			

My biggest surprise on this case:

ONE FOR THE ROAD — MY NEIGHBOURHOOD HUNT

Before Chapter 2, spot ONE machine near home that you now believe is a **learner**, not a rule-follower. What did it do that gave it away — and what data was it learning from?

2. Data: The Food of AI

GUIDING QUESTION

What do machines learn from — and what happens when the food is bad?

Machines cannot learn on empty stomachs — their food is **data**. You will hunt the four kinds of data around your classroom, tidy them into tables, turn a table into a chart, and then discover what happens when a learner is fed bad examples. Garbage in, garbage out — and only a detective can spot why.

MISSIONS T6-06 – T6-09
OUTCOMES 6.D1 · 6.D2 · 6.R2 **STRANDS** D

Links: Maths — tables & bar graphs · Science — observation

In this chapter

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2.2 Tables make data tidy 21
2.3 A picture is a grid of numbers 22
2.4 Garbage in, garbage out 24

WORDS YOU'LL MEET HERE

data **pixel** **tally** **garbage in, garbage out** **bar chart**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Every machine that learns has to be **fed**. What do you think your family's phone "eats" all day? Write your best guess — you will check it at the end of this chapter.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · ruler · colour pencils · 8 small things to sort

2.1 Every learner needs food

You have learned that machines find patterns in examples. But what exactly do they look at? **Data.** Data is the collected facts of the world — the numbers, words, pictures and sounds that we gather up and hand to a learner. Data is to AI what food is to your body: no learner can think without it, and a learner that eats bad food grows up weak and confused. A detective who understands data can predict exactly where an AI will do well — and where it will fall flat.

WORD POWER · 5

Definition: data

Collected facts about the world: numbers, words, pictures and sounds.

Read it three times today — it sticks!

DETECTIVE DRILL · WHICH KIND OF DATA?

The fact…	N	W	P	S
the temperature at 9 o'clock this morning				
your school anthem, sung in assembly				
a map of India on the classroom wall				
the words of your favourite poem				
a voice message from your grandmother				
the number of students absent today				

N = numbers · W = words · P = pictures · S = sounds. One row may fit two kinds — both ticks earn respect when you can say why.

TWO KINDS AT ONCE — THE TRICKY CASES

Real data rarely asks permission. A music video is **sounds and pictures** at the same time. A WhatsApp voice note with a transcript underneath is **sounds** turned into **words**. A traffic camera stores **numbers** that another program turns back into **pictures**. When you meet data that fits two boxes, do not panic — write both letters and say why. Machines that work with mixed data (a video app, a smart speaker, a self-driving cart) need detectives who can spot **every** kind hiding inside.

T6-06 Four Kinds of Data Hunt

QUICK WIN · PAIR · 10 MIN

Hunt your classroom (and your own pockets) for one example of each kind of data. Write or draw it in the box. The first pair to finish all four wins the round — but every box needs a real finding!

1 · NUMBERS

2 · WORDS

3 · PICTURES

4 · SOUNDS

One piece of data that could belong to TWO kinds at once:

2.2 Tables make data tidy

Loose data is like a scattered card pile: hard to think about. The first tool of every data detective is the **table** — rows and columns that line facts up neatly. Once data sits in a table, you can **tally** it (count with ||| strokes), and once you have tallies you can **draw a bar chart** that makes the biggest and smallest jump out at your eyes. Machines do exactly the same trick, only much faster: tidy first, then look.

DETECTIVE DRILL · READ THE TALLY

Tally strokes travel in groups of five: |||| with a strike-through. Write the number each tally stands for:

|||| = _____ || = _____ |||| |||| | = _____ |||| || = _____

T6-07 Table to Chart

GROUP · 15 MIN

In your group, ask each member: **“Which of these five do you like best?”** Add your teacher’s vote too if you like. Tally the answers, write the count, then colour one bar for each sport on the chart. Remember: one full tally is four strokes plus a strike-through.

Sport	Tally	Count
Cricket		
Football		
Badminton		
Kabaddi		
Other		



The tallest bar is … because …

One thing my chart proves, and one thing it cannot tell me:

2.3 A picture is a grid of numbers

Here is a secret that surprises almost everyone: to a camera and to an AI, your photograph is not a picture at all — it is a **huge grid of numbers**. Each tiny square of the image (a **pixel**) is stored as a number saying how bright or which colour it is. When you draw by numbers below, you are doing exactly what a camera does: turning a picture into numbers. Machines that “see” are really machines that are very good at finding patterns in those numbers.

0	1	1	1
1	0	0	0
1	0	0	0
0	1	1	1

A tiny 4 × 4 example. The zeros and ones store a letter — which one? Each row of the picture is one row of numbers. A real photo does this with millions of squares and numbers from 0 to 255 for brightness. **You** will speak "pixel language" in the next mission.

T6-08 Pixel Picture

PAIR · 15 MIN

How to play: Detective 1 keeps this page secret-ish and reads the code table aloud, row by row: "Row 1: 0 2 2 0 0 2 2 0 …". Detective 2 shades the left grid: 0 = leave white, 1 = shade light, 2 = shade dark. Then swap roles while Detective 1 shades the right grid. Finally reveal — do the two pictures match? That is data storage working!

0	2	2	0	0	2	2	0
2	1	1	2	2	1	1	2
2	1	1	1	1	1	1	2
2	1	1	1	1	1	1	2
0	2	1	1	1	1	2	0
0	0	2	1	1	2	0	0
0	0	0	2	2	0	0	0
0	0	0	0	0	0	0	0

CODE CARD

0 = white
1 = light
2 = dark

REVEALED! What did the numbers turn into? **A picture on a screen is really just** _____ stored in a grid.

Our pictures matched? ◯ **YES** ◯ **NO**

BONUS ROUND — INVENT A SECRET PICTURE

On scrap paper, design your own 8 × 8 secret picture and write its code card (rows of 0s, 1s and 2s). Swap code cards only — never the pictures — with another pair. If their shading matches your drawing, you have just proved the whole idea of this chapter: **the numbers are the picture.** A camera does this trick sixty times every second for every pixel it owns.

CAMERA-SPEAK — THREE WORDS DETECTIVES SHOULD KNOW

Pixel — one tiny square of the grid; the smallest piece of a picture a machine can see. **Resolution** — how many pixels the grid holds; more pixels = more numbers = more detail. **Brightness** — the number stored in each pixel, from 0 (pure black) to 255 (pure white). Next time a photo looks blurry on a screen, you will know what to blame: not enough pixels in the grid!

DETECTIVE DRILL · READ THIS CHART

The garden club asked 25 members: **"Which plant should we grow more of?"** Their data, tidied into a chart:

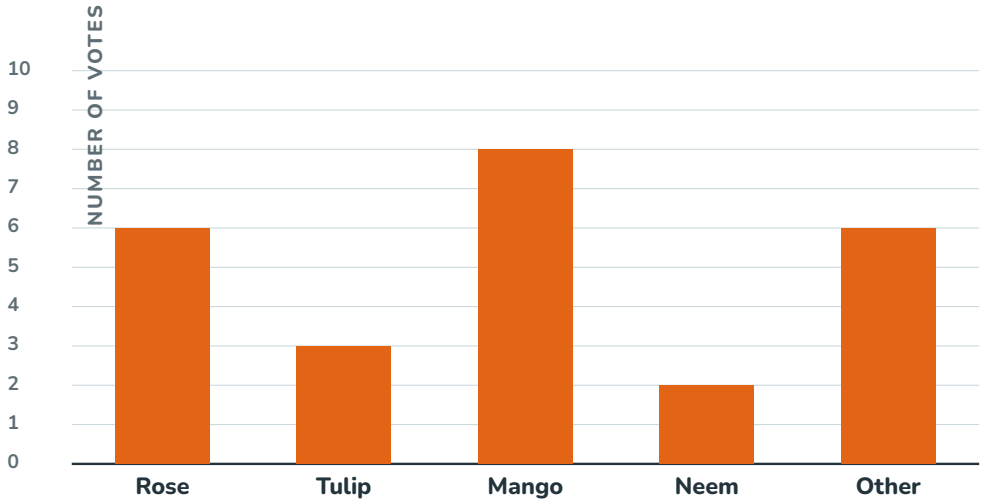


Chart question	My answer
Which plant won the vote? How do you know?	
How many more votes did Mango get than Tulip?	
A friend says "Tulip is the club's favourite plant." Is the chart on their side?	

2.4 Garbage in, garbage out

Now you can spot the biggest danger in an AI's kitchen. If the **training examples** are wrong, messy or one-sided, the machine cannot know it — it will happily learn a wrong pattern and then **predict** wrong answers forever, with total confidence. Detectives have a saying for this: **garbage in, garbage out**. It is the number one reason a smart machine says something silly — not because the machine is broken, but because its food was bad.

MYTH BUSTER

MYTH *"AI = computers = coding."*

TRUTH AI is an **idea** — learning patterns from data — and you can understand it fully with paper, as you just did. Not every computer runs AI (your calculator never learns), and AI is not the same as coding: coding writes fixed steps, while AI learns its own patterns from data.

T6-09 Garbage In, Garbage Out

GROUP · 10 MIN

Your group gets the eight **training cards** below (cut them out or copy them onto paper). One member is the **Machine**: cover the table, look only at the clues, and answer with the label that matches. The Machine announces its rule. Then the group tests the Machine with the three **test cards** at the bottom. Something will go wrong — good detectives find out why.

TRAINING CARDS — WHAT THE MACHINE SEES

Label on card	Clues on card
PET: CAT	whiskers · purrs · chases mice
PET: CAT	whiskers · sleeps a lot · soft fur
PET: DOG	barks · wags tail · fetches sticks
PET: CAT	barks · wags tail · fetches sticks
PET: COW	moos · eats grass · gives milk
PET: GOAT	bleats · eats everything · climbs
PET: HEN	clucks · lays eggs · scratches soil
PET: GOAT	moos · eats grass · gives milk

TEST CARDS — WHERE IT GETS EMBARRASSING

TEST 1
tiny puppy: barks · wags tail
Machine said: _____

TEST 2
kitten: whiskers · purrs
Machine said: _____

TEST 3
calf: moos · gives milk
Machine said: _____

The wrong rule our Machine learned:

Which training cards poisoned the learning? Circle them above. Why was the Machine fooled?

DETECTIVE'S NOTE

When an AI gives a wrong or unfair answer, do not just ask "what is wrong with the machine?" Ask the detective question: **"What is wrong with its data?"** One bad example in a million can still bend the pattern — that is why people must always stay in charge of checking.

CHAPTER 2 CLUES — SAY IT BACK

Data is the …

Garbage in, garbage out means …

A picture on a screen is really …

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can name the four kinds of data and point to one example of each in this classroom ○○○
- [ ] I can tidy data into a table, tally it, and draw a bar chart that others can read ○○○
- [ ] I can explain how a picture becomes a grid of numbers — and show it with my own code ○○○
- [ ] I can explain “garbage in, garbage out” to someone who missed today’s class ○○○

THINK & INK

A machine I know eats data all day long. The data it probably eats is … and if that data is messy, it will …

MY CASE JOURNAL — THE CLUE I WILL NEVER FORGET FROM THIS CHAPTER

3. Patterns & Decisions

GUIDING QUESTION

How does a machine decide what to do next?

Patterns are the clues learners chase: find one, predict what comes next, and then say **how sure** you are. You will hunt number and letter patterns, play Guess My Rule with a hidden rule-keeper, trace an if-then decision machine, and debate answers whose confidence is higher than their accuracy.

MISSIONS T6-10 – T6-13
OUTCOMES 6.L1 · 6.T1 · 6.R2 **STRANDS** L T

Links: Maths — number patterns · English — clear questions

In this chapter

3.1 Patterns are clues 28
3.2 Guess my rule 29
3.3 If-then: the question tree 31
3.4 How sure am I? 32
3.5 Chapter 3 wrap-up — my case summary 34

WORDS YOU'LL MEET HERE

pattern · prediction · confidence · if-then · question tree

WARM-UP · BEFORE YOU READ — ALONE · 2 MIN

Clap a secret rhythm three times, then stop. Can your partner continue it? That is a pattern doing its job — write the rhythm you clapped, using letters like L-S-L-S.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · scrap paper for secret rules · a partner who can keep a secret

3.1 Patterns are clues

A detective's sharpest tool is not a magnifying glass — it is **noticing**. A pattern is something that repeats in a way you can spot and use: day after night after day, 2-4-6-8, the same chorus after each verse. Machines live on patterns. A learning machine studies its **training examples**, and every **pattern** it finds becomes a rule it can use to make a **prediction** — a smart guess about what comes next. Notice the order: first the pattern, then the prediction, then a guess about how sure it is. You will do all three in the next mission, exactly like a machine, on paper.

WORD POWER · 6

Definition: **pattern**

Something that repeats in a way we can spot, describe and use.

Read it three times today — it sticks!

WORD POWER · 7

Definition: **prediction**

A smart guess about what comes next, based on a pattern.

Read it three times today — it sticks!

DETECTIVE DRILL · PATTERN OR COINCIDENCE?

Not every repetition is a pattern worth predicting with. A **pattern** keeps working on new cases; a **coincidence** happens once or twice and then betrays you. Mark each one P (pattern) or C (coincidence) — and be ready to defend your choice:

The clue	P or C?	My defence in one line
The sun has risen every morning in recorded history.		
My bus was late on Monday and on Tuesday.		
Water always boils sooner on a mountain than at the seaside.		
I got two greens in a row on the first throws of a dice.		

*Machines make this exact mistake — they trust a repetition that was really a coincidence, because they cannot **wonder** whether it will keep working. Detectives can.*

T6-10 Pattern Hunt

ALONE · 10 MIN

For each strip: write what comes next, then shade how sure you are (one circle = a wild guess, two = fairly sure, three = I could defend it in court).

The clues	My prediction	How sure?
2 · 4 · 6 · 8 · ?		○○○
A · B · A · B · A · ?		○○○
● ■ ● ■ ● ■ ?		○○○
M · T · W · T · ? (days of the week!)		○○○
1 · 2 · 4 · 8 · ? (beware — this one doubles!)		○○○
J · F · M · A · ? (months!)		○○○

Which pattern fooled the most detectives in your class? Why?

A pattern from my own life (clap rhythm, bus timings, cricket over…):

MYTH BUSTER

MYTH *"AI knows everything — it is always right."*

TRUTH A machine can only **predict** from patterns in the data it was given. If its data missed something, or was one-sided, it will make a wrong prediction — and still say it confidently. "Confident" and "correct" are two different words, and detectives never confuse them.

3.2 Guess my rule

Here is a party trick that is secretly machine learning. One player thinks of a secret rule — say, "only even numbers may pass". The others **test** the rule by offering numbers, and the rule-keeper answers only YES or NO. Slowly, from these **labelled examples**, the players work out the hidden pattern. That is exactly how a learning machine trains: test, answer, adjust, test again. Play at least two rounds with different rule-keepers.

T6-11 Guess My Rule

GROUP · 15 MIN

One detective keeps a secret rule (examples your teacher may offer: **only even numbers · numbers bigger than 20 · numbers with a 3 in them · the letter must be a vowel**). Everyone else offers tests and records them below. When you think you know it, write the rule and let the rule-keeper grade you with a tick or cross.

Example I offered	YES (allowed) or NO (not allowed)

Our final guess at the rule:

How many tests did we need before we were sure? Did any test mislead us?

ROUND TWO — A HARDER RULE

Example I offered	YES or NO	Example I offered	YES or NO

Round-two rule, in one if-then sentence:

DETECTIVE'S NOTE

Did you notice? You never **told** the rule-keeper's rule to the group — they learned it from labelled examples, just like a machine. And just like a machine, you might have learned a rule that is **nearly** right. More tests = more confidence.

DETECTIVE DRILL · TRACE THE MACHINE

This little machine sorts numbers using an exact **algorithm**. Trace each test number through it, step by step, and write what the machine answers.

THE MACHINE'S ALGORITHM

1. Look at the number.
2. **If** it is even, answer "E" and stop.
3. **If** it is odd **and** bigger than 10, answer "OB".
4. Otherwise, answer "OS".

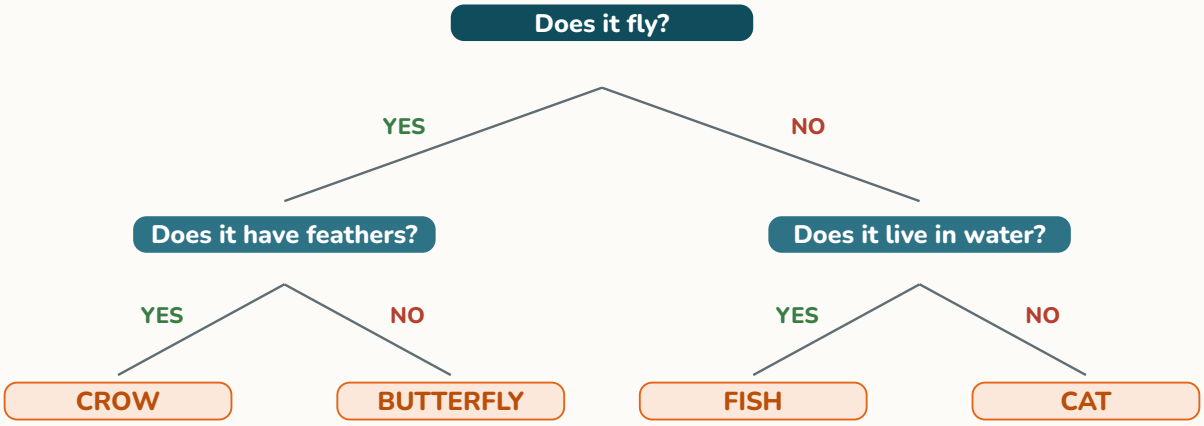
TEST IT WITH…

Test number	Machine's answer
7	
12	
23	
48	
9	

Which question should the machine ask FIRST to finish fastest — even, or bigger-than-10? Why?

3.3 If-then: the question tree

When a machine must **decide**, it often follows a chain of if-then questions: **if** the answer to Question 1 is yes, go left; **if** no, go right. You already use if-then thinking every day: **if** it is raining, take the umbrella; **if** the light is red, wait. People draw this chain as a **question tree**, and playing "20 Questions" is exactly this — each good question splits the possibilities in half. A question tree is also a kind of **algorithm**: exact steps that anyone — human or machine — can follow to the same answer. The better each question splits the suspects, the shorter the tree.



T6-12 20-Questions Tree

PAIR · 15 MIN

Your six suspects: **cow · crow · fish · snake · cat · frog**. Draw a yes/no question tree that tells all six apart. Start every branch from one top question. The pair whose tree needs the **fewest** questions on average wins the round.

DRAW YOUR QUESTION TREE HERE

Animal	Questions needed	Animal	Questions needed
Cow		Snake	
Crow		Cat	
Fish		Frog	

Our best first question (it split the suspects most evenly):

A question that turned out to be useless, and why:

3.4 How sure am I?

Real AI systems never just answer — they answer with a **confidence**: “cat, 62% sure”. Sometimes a machine is very confident and very wrong; sometimes it is unsure and right. People fall for the same trap: we shout answers we are proud of, and whisper the ones we doubt.

Detectives practise **saying how sure they are**, out loud, every single time. It is the thinking habit that keeps humans — not machines — in charge of the judging.

T6-13 Sure or Not Sure?

CLASS · 10 MIN

Your teacher reads six machine-style answers. For each one, mark its number on the confidence line where you think it belongs. Then the class debates the two that everyone placed differently.

#	The machine says…	Its claimed confidence
1	“The next number in 2-4-6-8 is 10.”	95%
2	“This photo has a cat in it.”	62%
3	“Tomorrow’s highest temperature: 31°C.”	78%
4	“Your name starts with the letter A.”	40%
5	“This email is spam.”	99%
6	“‘Recieve’ is the correct spelling.”	97%
7	“Your class will have a test on Thursday.”	55%
8	“The sun will rise tomorrow morning.”	100%

Write each statement number where its confidence belongs →



0% — total guess 50% 100% — certain

Statement 6 was 97% confident — and wrong! One reason a confident answer can still be wrong:

CLASS DEBATE — THE TWO WE PLACED DIFFERENTLY

Statement numbers we argued about: _____ and _____ The argument was worth it because… _____

THINK & INK

When my first prediction today was wrong, the real reason was … Next time before I answer, I will check by …

3.5 Chapter 3 wrap-up — my case summary

MY PATTERN GALLERY — PATTERNS I NOTICED THIS WEEK

A PATTERN FROM NATURE OR ART

A PATTERN FROM HOME OR SCHOOL

A pattern I found and the prediction it let me make:

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can find a pattern, predict what comes next, and say how sure I am ○○○
- [ ] I can test a hidden rule with examples until I am confident — like a machine trains ○○○
- [ ] I can draw a yes/no question tree that another pair can follow without my help ○○○
- [ ] I can explain why a confident answer is not always a correct answer ○○○

CHAPTER 3 CLUES — SAY IT BACK

A pattern is …

A prediction is …

Before I answer any question, I will say how …

Chapter 3 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Find the next item and name the pattern: 81 · 68 · 55 · 42 · ?

2. Draw or describe a two-question tree that separates **dog / bird / fish**.

3. A weather app says “rain, 55% sure.” What should you do — and what should you **not** believe?

MY CASE JOURNAL — THREE LINES ABOUT PATTERNS, PREDICTIONS AND HONESTY

4. Be a Smart Digital Citizen

GUIDING QUESTION

How do I enjoy the online world — and stay safe, kind and honest?

Smart detectives stay safe, kind and honest — online as everywhere. You will sort personal information with a privacy traffic light, cook unguessable practice passphrases, run the three-check Real-or-Fake detector on suspect cards, and follow one drawing's digital footprint as it travels the internet.

MISSIONS T6-14 – T6-17
OUTCOMES 6.R1 · 6.R2 **STRANDS** R

Links: Social Science — community life · English — checking facts

In this chapter

4.1 Privacy: my information, my choice 37
4.2 Passphrases: long and strange wins 38
4.3 Real or fake? Be the detector 40
4.4 Your digital footprint 41

WORDS YOU'LL MEET HERE

privacy **personal information** **passphrase** **verify** **digital footprint**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A message pops up: "You have WON a free phone! Click here in the next 10 minutes!" How does it try to make you feel — and what would a detective do first?

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · colours for the traffic light · a younger student to teach (later)

4.1 Privacy: my information, my choice

Every time you use a game, an app or a website, somebody is collecting **data** — and some of that data is **about you**. Your name, your school, your street, your photos, even your jokes: this is **personal information**, and you get to decide who may see it. **Privacy** means keeping personal things shared only on purpose, with people you trust. Machines that learn from data cannot judge what **should** stay private — that judgement is a human job, and from today it is yours.

WORD POWER · 8

Definition: privacy

Keeping personal information shared only on purpose, with people you trust.

Read it three times today — it sticks!

T6-14 Privacy Traffic Light

ALONE · 10 MIN

Sort the ten items into three lights. GREEN = okay to share freely · YELLOW = ask a grown-up first · RED = never share with apps, strangers or websites. Write each item's number in the right column.

The ten items

1 my favourite colour · 2 my full name and school · 3 a photo of my home street · 4 my favourite game · 5 my phone number · 6 a drawing I made of a cat · 7 my home address · 8 my best friend's secret · 9 my password · 10 my class timetable

GREEN	YELLOW	RED
share freely	ask a grown-up first	never share

One item was hard to place, because...

DETECTIVE'S NOTE

Strangers online are exactly like strangers anywhere: mostly ordinary people, sometimes not. The rule does not change — personal information goes only to people your grown-ups trust, no matter how friendly a screen looks.

4.2 Passphrases: long and strange wins

A short password like `riya123` is cracked in seconds by a machine that simply tries every combination — that machine is doing **automation**, fixed steps, millions per minute. What defeats it is a **passphrase**: three or more **unrelated** words glued together, long enough that guessing would take centuries. The strange words do not need to make sense — they need to be **long, unrelated to you, and known only by you**.

DETECTIVE DRILL · HOW LONG WOULD A CRACKING MACHINE NEED?

A cracking machine tries millions of guesses per minute. Security experts estimate the **crack time** — how long a machine would need to try every combination. Read the estimates, then rank the four passwords from weakest to strongest (1 = weakest):

The password	Estimated time to crack	My rank 1–4
riya123	less than a second	
Riya@2015	about 2 months	
Green-Onion-Tiger-77?	about 200 years	
Rickshaw-Monsoon-Pencil-42!	centuries	

*What made the difference? **Length** first, **unrelated words** second, **numbers and symbols** third. Notice that Riya@2015 has a symbol and numbers too — but it is built from a real name and a year, so a machine guesses it early.*

DETECTIVE DRILL · THE STRANGE-WORD GENERATOR

Strong passphrases start with **unrelated** words. Practise mixing: take the last letter of your name, the first fruit you saw today, and the name of a street you have walked — glue them with a number and a symbol. Do it twice below; the stranger it looks, the better it works:

Mix 1: **Mix 2:**

T6-15 Password Recipe

PAIR · 10 MIN

Cook up a **practice** passphrase using the recipe: pick one word from each column, add your favourite two-digit number and a symbol. Example: “Rickshaw-Monsoon-Pencil-42!”. Write it, then rate it. **Never write your real password anywhere — and never show this page’s passphrase to anyone outside your pair.**

Pick one from each column	A	B	C
	Rickshaw	Monsoon	Pencil
	Cricket	Thunder	Mango
	Peacock	Banyan	Lantern
	Kite	Pepper	Compass

Our practice passphrase:

Two reasons it is strong:

Too weak: names, birthdays, phone numbers, “123456”, your favourite team — machines guess these first.

Strong: 12+ characters, unrelated words, mixed with numbers and symbols, changed if ever shared.

DETECTIVE DRILL · RATE THESE TWO PRACTICE PASSPHRASES

The practice passphrase	Strong or weak?	Why?
riya123		
Rickshaw-Monsoon-Pencil-42!		
AJAY2015		
Green-Onion-Tiger-77?		

DETECTIVE DRILL · PASSPHRASE GYM — BUILD THREE, RATE EACH

Cook three **practice** passphrases with the recipe (one word from each column + two digits + a symbol). Make one easy to remember but still long, one completely strange, one for a make-believe account. Rate each honestly:

Practice passphrase	Length (count!)	Rate it /5	One thing that makes it strong

*Remember the recipe: **long** beats **clever**. "Rickshaw-Monsoon-Pencil-42!" takes a cracking machine centuries; "Riya@123" takes it seconds. Length is the superpower.*

4.3 Real or fake? Be the detector

The online world is full of edited photos, forwarded rumours and — increasingly — pictures and text made by AI. None of this means the internet is bad; it means every good detective needs a checking routine. Professionals call it **verification**, and it needs only three questions: **Where is it from? (source) · What is the evidence? · How does it try to make me feel?** If a message makes your heart race — fear, anger, "forward this NOW!" — that rushing feeling is exactly when you should slow down and verify.

WORD POWER · 9

Definition: **verify**

To check carefully whether something is true, using evidence and trusted sources.

Read it three times today — it sticks!

DETECTIVE DRILL · SPOT THE EMOTION WORDS

Strong feeling is the costume that fake messages wear. Underline the emotion words in each suspect message, then write what feeling it aims at — and what a detective does first:

The suspect message	Feeling it aims at	What a detective does first
"SHOCKING!!! Doctors HATE this one strange trick…"		
"Forward NOW or your family will regret it!"		
"A calm report: the new library opens on Tuesday."		

*Notice the last row: true information usually **whispers**. If a message shouts, slow down — that is exactly the moment to verify.*

DETECTIVE'S NOTE — AI ANSWERS CAN BE WRONG

An AI answer can be wrong for a simple reason: the machine **predicts** from patterns in its training data, it does not **know** things the way people do. If its data was one-sided, messy or old, its answer will show it. One way to check any answer — human or machine: **verify with a second source** before you believe or forward it.

T6-16 Real or Fake Detective

GROUP · 15 MIN

Your teacher shows three printed cards (or use these three suspects). Apply the three checks to each: **source · evidence · emotion**. Give a verdict: REAL, FAKE, or NOT SURE — and "not sure" is an honest detective's answer when evidence is missing!

CARD A
A photo of a dog — but it has five legs and very odd fur. Forwarded with "Amazing new breed!".

CARD B
"Forward this to 10 friends in 10 minutes or your account will be closed tomorrow!!!"

CARD C
A poster claims "Scientists say our city's air is the cleanest ever" — with no scientist's name, no date, no source.

Card	Our verdict (REAL / FAKE / NOT SURE)	Which check helped most?
A		
B		
C		

One thing on this page that a machine might believe but a detective should doubt:

DESIGN ONE MORE SUSPECT CARD — SWAP WITH ANOTHER PAIR AND DETECT THEIRS!

4.4 Your digital footprint

Imagine walking on a beach of soft sand: every step leaves a mark, and some marks stay long after you have gone. Everything you post, like, share or search leaves a mark too — a **digital footprint**. Footprints are not automatically bad: your kind comments and your good work can travel far and make people smile. But a footprint is hard to erase — one screenshot, one forward, and a moment you meant for two friends can be standing in front of a whole school. Post like the whole world is watching, because one day it might be.

Definition: **digital footprint**

The trail of marks you leave behind when you post, like, share or search online.

Read it three times today — it sticks!

T6-17 Footprint Walk

ALONE · 10 MIN

Follow one imaginary post — "You post a drawing for the school contest" — and watch it travel. For each hop, note **who can see it now** and whether you could still take it back.

Hop	Who can see my drawing now?	Can I take it back? (YES / NO)
1 · I post it for the contest		
2 · My class group forwards it		
3 · My teacher saves it for the website		
4 · The school website shows it		
5 · A cousin shares it to family far away		
6 · Unknown accounts copy it		
7 · A search engine indexes it		

One habit that keeps a footprint kind and small:

My footprint promise this month:

DETECTIVE DRILL · WRITE THE KIND REPLY

A classmate posts one of these in the class group. What is a kind, helpful reply? Write one — it must pass the T.H.I.N.K. check!

The post…	My kind reply
"I came last in the spelling test. I am so bad at everything."	
"Look at this photo I took of our school garden!"	
"Nobody wants to be my science partner."	

Before you post — run the T.H.I.N.K. check

- **T** Is it True?
- **H** Is it Helpful?
- **I** Is it Inspiring?
- **N** Is it Necessary?
- **K** Is it Kind?

If any answer is NO — hold the post, detective. Screens can be big; kindness is bigger.

MY FIVE SAFETY HABITS — SAY THEM OUT LOUD

1. Private info stays private
2. Long, strange passphrases
3. Think before posting
4. Be kind — always
5. Check before believing

MYTH BUSTER

MYTH *"Bigger screen = smarter machine."*

TRUTH Smartness comes from **data and patterns**, not screen size. A tiny speaker can run a powerful learner, and a giant TV can be completely ordinary. Judge a machine by what it **learned from** — never by how big it looks.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can sort personal information into share / ask-first / never-share, and explain why ○○○
- [ ] I can build a strong practice passphrase — and I know never to write my real one down ○○○
- [ ] I can run the three checks (source · evidence · emotion) on anything doubtful ○○○
- [ ] I can explain my digital footprint to a younger student in two sentences ○○○
- [ ] I use the T.H.I.N.K. check before I post — and I can prove it ○○○

THINK & INK

One habit I will start this week is … because last week I saw / did …

Chapter 4 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Which of these is okay to type into a website that asks: your full name and school, or your favourite colour? Why?

2. Turn “mango”, “monsoon”, “7” and “!” into a strong passphrase — and say what makes it strong.

3. A forwarded message shouts “SHARE IMMEDIATELY!!!” What are the THREE checks you run before deciding?

MY CASE JOURNAL — THE SAFETY HABIT I WILL NEVER SKIP

5. My Neighbourhood AI Investigation

GUIDING QUESTION

What can I discover about AI near me — using my own data?

Your biggest case yet: a **real survey** of five grown-ups about the smart machines around them. You will plan and run the survey, organise and chart the data you collect, design a helper machine of your own — and then close the case by answering the big question from page one, with evidence.

MISSIONS T6-18 – T6-20
OUTCOMES 6.U1 · 6.D1 · 6.W1 · 6.R2 **STRANDS** U D L W R

Links: Social Science — community survey · Maths — data & charts

In this chapter

5.1 Run a real survey 46
5.2 Design a helper machine 48
5.3 Close the case 49

WORDS YOU'LL MEET HERE

survey · tally · bar chart · limit · evidence

WARM-UP · BEFORE YOU READ — ALONE · 2 MIN

Practise the survey question out loud once: "Where do you meet machines that seem smart in your daily life?" Now write one answer you might expect to hear — you will check it against your real data soon.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · the survey page · five kind grown-ups · your best listening ears

YOUR BIGGEST CASE YET — READ THIS FIRST!

Every skill from Chapters 1–4 comes together here. In groups, you will run a **real survey** with five grown-ups, organise the **data** you collect, draw it, and present findings — then design a helper machine of your own. It takes 2–3 class periods plus home time. Detectives work carefully: one sloppy column can spoil a whole investigation.

5.1 Run a real survey

GROUP ROLES — DECIDE BEFORE YOU GO

Role	Detective
Asker — asks the questions politely	
Recorder — writes the answers down	
Checker — re-reads each row for mistakes	
Presenter — shares the findings with the class	

SURVEY MANNERS — READ BEFORE YOU KNOCK

Ask first: "Do you have two minutes to help our class project?" **Listen** without interrupting — a rush recorder misses data. **Write initials only** — never a neighbour's full name. **Say thank you**, even when the answer is "no". **No photos, no recordings, no personal details** — our survey is paper-only.

DETECTIVE DRILL · PREDICT, THEN GO AND CHECK

Before real detectives collect data, they write a **prediction** — then the data gets a vote. What do you predict your five adults will name most often? Circle one, then race to see if your pattern-holding skills from Chapter 3 were right:

○ Phone & video apps ○ Voice assistants ○ Shops & payments ○ Travel & traffic ○ Somewhere else

A prediction that survives contact with real data earns a detective's smile. One that fails teaches you something the easy way.

GROUP LOGISTICS · RECRUIT YOUR FIVE

Plan like professionals: agree **who** your group will ask (roles, never full names in print) and **when**. A plan made today saves a wasted trip tomorrow:

Investigator to ask (role)	Where we will meet them	Best day and time

T6-18 AI in My Neighbourhood — the survey

GROUP · 2–3 periods MIN

Step 1 · Plan. As a group, agree whom you will ask: five adults — a family member, a neighbour, a shopkeeper, a teacher, anyone safe and willing. The question: **"Where do you meet machines that seem smart in your daily life?"** Write your plan before you collect:

We will ask (roles, not names):

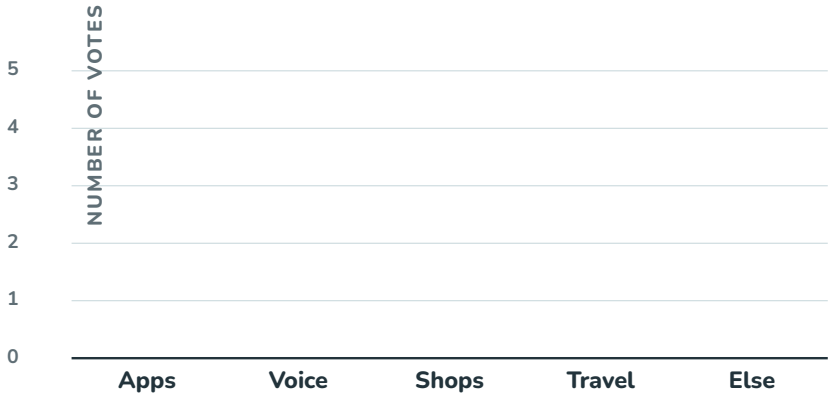
We will collect our data on (day/place), so the survey is safe and polite because…

Step 2 · Collect. Record one line per person. Ask the follow-up: **"What does it probably learn from?"** — that is its data.

Who (initials only)	The smart machine they named	What data might it use?	Rule-follower or learner?

Step 3 · Organise. Tally your group's five answers into the families below, then draw the bars.

Family	Tally
Phone & video apps	
Voice assistants	
Shops, banks & payments	
Travel & traffic	
Somewhere else	



Step 4 · Findings — finish the sentences like a real analyst:

5.2 Design a helper machine

T6-19 My Helper Machine — the poster

GROUP · 2 periods MIN

Design (on paper!) one machine that would genuinely help people you know. Fill the case sheet, then make a poster your class can judge.

MACHINE NAME

THE PROBLEM IT SOLVES (who struggles, with what?)

WHO WILL USE IT

ONE HONEST LIMIT — "it cannot … because …"

- ☐ **RULE-FOLLOWER — it runs an algorithm we write: exact if-then steps, same every time.**
- ☐ **LEARNER — it studies training examples and finds its own pattern.**

Tick one. Every helper machine is one of these two — choosing well is the designer's first job.

WHAT DATA DOES IT EAT? TICK ALL THAT APPLY

- ☐ Numbers — prices, times, counts
- ☐ Words — names, questions, messages
- ☐ Pictures — photos, drawings, maps
- ☐ Sounds — voice, music, alarms

SKETCH YOUR HELPER MACHINE — LABEL THE DATA GOING IN AND THE HELP COMING OUT

HOW YOUR POSTER WILL BE JUDGED — REASONING EARNS MORE THAN ART

Level	What it looks like
Beginning	A machine is named, but the problem, data or limit is missing.
Growing	The problem and data are clear, but the limit is missing — every real machine has one!
Secure	Problem, users, data and an honest limit all make sense together.
Shining	Everything in Secure, plus the group defended their choices with evidence from their survey — and said where the machine could fail.

5.3 Close the case

T6-20 I used to think... Now I think...

ALONE · 10 MIN

The last mission of the case: look back at page 1's big question and at everything your pencil went through.

I USED TO THINK

NOW I THINK

Three questions I still have for my next case (Class 7):

★ **Detective badge earned: LEVEL 1 · NOTICE — you can now see where AI is, and tell it from ordinary machines.**

CASE CLOSED

A LETTER TO THE CLASS 7 DETECTIVE — WHO WILL BE YOU

Next year a new case opens: **"How can a machine learn without being told the rules?"** Leave three clues here for your future self — one thing you want to remember, one thing you want to understand better, and one question you dare Class 7 to answer:

My AI words

The ten words on this case — each one met through an experience, never before.

1 algorithm
An exact list of steps that tells a machine how to do a task, one step at a time.
USE IT IN MY OWN SENTENCE:

2 automation
Work done by a machine that repeats fixed steps in the same way every time.
USE IT IN MY OWN SENTENCE:

3 artificial intelligence
Technology that lets machines do tasks that seem to need human intelligence.
USE IT IN MY OWN SENTENCE:

4 training examples
The examples we show a machine so that it can find patterns by itself.
USE IT IN MY OWN SENTENCE:

5 data
Collected facts about the world: numbers, words, pictures and sounds.
USE IT IN MY OWN SENTENCE:

6 pattern
Something that repeats in a way we can spot, describe and use.
USE IT IN MY OWN SENTENCE:

7 prediction
A smart guess about what comes next, based on a pattern.
USE IT IN MY OWN SENTENCE:

8 privacy
Keeping personal information shared only on purpose, with people you trust.
USE IT IN MY OWN SENTENCE:

9 verify
To check carefully whether something is true, using evidence and trusted sources.
USE IT IN MY OWN SENTENCE:

10 digital footprint
The trail of marks you leave behind when you post, like, share or search online.
USE IT IN MY OWN SENTENCE:

DETECTIVE'S OATH

Machines **predict, match, estimate** — people **design, choose data and stay responsible**. When a machine speaks, I will ask my three questions: **How do I know? What is missing? Who made this?**

A NOTE FROM THE WORD KEEPER

Words drift as machines change: **computer** once meant a person who computes, and **cloud** moved from the sky into your pocket. Do not be surprised if the words in your dictionary mean something slightly different in ten years — a detective keeps watching, and keeps asking.

My detective's dictionary

The ten words above were planted for you. These pages are for the words you found on your own.

11
WHAT I THINK IT MEANS:
WHERE I MET IT:

12
WHAT I THINK IT MEANS:
WHERE I MET IT:

13
WHAT I THINK IT MEANS:
WHERE I MET IT:

14
WHAT I THINK IT MEANS:
WHERE I MET IT:

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WHAT I THINK IT MEANS:
WHERE I MET IT:

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WHAT I THINK IT MEANS:
WHERE I MET IT:

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WHAT I THINK IT MEANS:
WHERE I MET IT:

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WHAT I THINK IT MEANS:
WHERE I MET IT:

19
WHAT I THINK IT MEANS:
WHERE I MET IT:

20
WHAT I THINK IT MEANS:
WHERE I MET IT:

THE DETECTIVE'S DICTIONARY HABIT — CHECK A NEW WORD THREE WAYS

When you write a new word above, test it before you trust it: **1** use it in a sentence of your own and read it out loud — does it sound right? **2** explain it to somebody younger — if they nod, you own it. **3** look where the word came from if you can — a word with a visible history is a word you never forget. Three checks, and the word moves into your permanent kit.

WORD HUNT — WHERE DETECTIVES COLLECT NEW WORDS

A new AI word can come from anywhere: a news headline overheard at breakfast, a phone app that asks for your **permission**, a grown-up saying **algorithm** at the dinner table, a sign at the railway station. When you hear a word you cannot explain yet — that is not a failure, that is a **case opening**. Write it here, ask around, and check it against this book's ten words: does it change what they mean, or add a new room to the house?

My progress

Shade honestly — detectives never fake evidence.

MY OUTCOME TRACKER — TICK WHEN YOU TRULY CAN DO IT

- ☐ 6.U1 — say what AI is, and tell a learner from a fixed-rule machine
- ☐ 6.U2 — name two human strengths and two machine strengths, with reasons
- ☐ 6.D1 — name the four kinds of data; sort a table; draw a bar chart
- ☐ 6.D2 — show how a simple picture becomes numbers on a grid
- ☐ 6.L1 — find a pattern, predict the next step, say how sure I am
- ☐ 6.L2 — describe the three ways of learning with human stories
- ☐ 6.W1 — point to AI at home, school and neighbourhood, and name its data
- ☐ 6.R1 — use my five safety habits — every day, not just today
- ☐ 6.R2 — explain why an AI answer can be wrong, and how to check it
- ☐ 6.T1 — break a task into exact steps another person can follow

MY CASE FILES — SHADE A STAR PER CHAPTER FINISHED

1 Machines · 2 Data · 3 Patterns · 4 Digital Citizen · 5 Capstone

☆ ☆ ☆ ☆ ☆

PRATIMAI · AI HANDOUTS · CLASS 6

My AI Detective Certificate

This certifies that detective

_____

DETECTIVE NAME

has completed the case file **"AI Around Me"** — Level 1 · NOTICE — solved all twenty missions, and promised to keep asking: **How do I know? · What is missing? · Who made this?**

_____ _____ _____

DATE | DETECTIVE'S SIGNATURE | GUIDE'S SIGNATURE

Ready for the next case? Class 7 asks: **"How can a machine learn without being told the rules?"** — bring your badge.